

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1-2

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1. TITLE

Develop a Styled Webpage Using HTML5 Canvas and CSS3

2. AIM

To design a styled webpage using HTML5 Canvas and CSS3, demonstrating canvas drawing, navigation, cards, buttons, gradients, borders, rounded corners, shadows and a footer.

3. OBJECTIVE

The objectives of this experiment are:

- To create a webpage with a structured header, navigation bar and footer.
 - To use CSS3 colors, gradients, spacing, borders, rounded corners and shadows.
 - To create and style an HTML5 canvas element.
 - To draw a rectangle, circle, line and text using JavaScript canvas methods.
 - To arrange three information cards using CSS Flexbox.
 - To demonstrate styled buttons and hover effects.
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4. THEORY

The program combines HTML, CSS3 and JavaScript. HTML provides the page structure through the header, navigation, canvas section, cards and footer. CSS3 provides the visual styling through a linear gradient background, dark header and navigation, text shadows, borders, rounded corners, box shadows, spacing and button hover effects. The JavaScript code accesses the canvas using `getElementById()`, obtains the 2D drawing context, and uses `fillRect()`, `arc()`, `lineTo()`, `stroke()` and `fillText()` to draw the displayed graphics and the text "vijayasri"

5. ALGORITHM / PROCEDURE

1. Create the HTML document and add the title.
2. Apply the general CSS reset and gradient page background.
3. Style the header, navigation links and hover behavior.
4. Create the HTML5 canvas section and apply a border and shadow.
5. Create three cards using a flex container and style their borders, padding, rounded corners and shadows.
6. Add styled buttons and define their hover effect.
7. Add the footer with matching styling.
8. Create a JavaScript canvas context and draw the rectangle, circle, line and text.
9. Save the file as code.html and open it in a browser.
10. Verify the rendered canvas, cards, buttons and footer using the provided output screenshot.

6. PROGRAM

code.html

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Styled Webpage</title>

<style> /* ===== General Styling
===== */ *{ margin:0; padding:0;
box-sizing:border-box;

}

body{ font-family:Arial, sans-serif;
background:linear-gradient(to right,#74ebd5,#9face6);
color:#333;
} /* ===== Header ===== */ header{
background:#2c3e50; color:white;
text-align:center; padding:20px;
text-shadow:2px 2px 5px black;

}

/* ===== Navigation ===== */
nav{ background:#34495e;
padding:15px;
text-align:center;
}

nav a{ color:white;
text-decoration:none;
margin:20px;
font-size:18px;
font-weight:bold;
transition:0.3s;
```

```

nav a:hover{
  color:yellow;
  background:#1abc9c;
  padding:8px 15px;
  border-radius:5px;
} /* ===== Canvas Section ===== */ .canvas-section{

```

```

  text-align:center;
  margin:30px;
}

```

```

canvas{
  border:3px solid black;
  background:white;
  box-shadow:5px 5px 10px gray;
}

```

```

/* ===== Cards ===== */

```

```

.container{
  display:flex;
  justify-content:center;
  gap:20px;
  flex-wrap:wrap;
  margin:30px;
}

```

```

.card{
  width:280px;
  background:white;
  padding:20px; border:2px
  solid #3498db;
  border-radius:15px;
  box-shadow:5px 5px 10px gray;
  text-align:center;
} .card

```

```

h3{
  font-family:Georgia;
  color:#2c3e50;
}

```

```

.card p{
  margin-top:10px;
  font-size:16px;
}

```

```

button{
  margin-top:15px;
  padding:10px 20px;
  border:none;
  border-radius:10px;
  background:#3498db;
  color:white;
  font-size:16px;
  cursor:pointer;
  transition:0.3s;
}

```

```

button:hover{
  background:#2ecc71;
  transform:scale(1.1);
} /* ===== Footer ===== */ footer{

```

```

  background:#2c3e50;
  color:white;
  text-align:center;
  padding:15px; margin-top:20px;
  text-shadow:2px 2px 5px black;
}

```

```

</style>

```

```

</head>

```

```

<body>

```

```

<header>

```

```

  <h1>My Styled Webpage</h1>
  <p>HTML5 Canvas and CSS3 Styling
</header>

```

```

<nav>

```

```

<a href="#">Home</a> <a
href="#">About</a> <a
href="#">Services</a> <a
href="#">Contact</a>
</nav> <section
class="canvas-section">

<h2>HTML5 Canvas</h2>
<br>
<canvas id="myCanvas" width="600" height="300"></canvas>

</section>

<section class="container">

<div class="card"> <h3>Card One</h3>

</div>

<p>This card demonstrates borders, padding, margins, rounded corners and shadow effects.</p>
<button>Read More</button>

<div class="card"> <h3>Card Two</h3> </div>

<div class="card"> <h3>Card Three</h3> </div>

<p>CSS3 makes webpages attractive using gradients, hover effects, fonts and colors.</p> <button>Learn
More</button>

<p>HTML5 Canvas allows drawing graphics directly inside a webpage using JavaScript.</p>
<button>Explore</button>

</section>

<footer>
© 2026 My Styled Webpage | All Rights Reserved
</footer>

<script>

var canvas = document.getElementById("myCanvas"); var ctx =
canvas.getContext("2d");

/* Rectangle */
ctx.fillStyle="blue";
ctx.fillRect(40,40,150,80);

/* Circle */ ctx.beginPath();
ctx.arc(300,80,40,0,2*Math.PI);
ctx.fillStyle="red"; ctx.fill
();

/* Line */ ctx.beginPath
(); ctx.moveTo(420,40);
ctx.lineTo(550,150);
ctx.strokeStyle="green";
ctx.lineWidth=4;
ctx.stroke();

/* Name Text */
ctx.fillStyle="purple";
ctx.fillText("vijayasri",220,220);
ctx.font="28px Arial";

</script>

</body>
</html>

```

7. CODE EXPLANATION

Code	Explanation
body	Sets the Arial font, gradient background and text color.
header	Styles the heading area with dark background, white text and shadow.

Code	Explanation
nav / nav a	Creates the navigation bar and styled navigation links.
.canvas-section	Centers the canvas section and adds spacing.
canvas	Adds a black border, white background and shadow to the canvas.
.container / .card	Arranges and styles the three cards using Flexbox.
button / button:hover	Styles the buttons and changes their appearance on hover.
getContext("2d")	Obtains the 2D drawing context for the canvas.
fillRect() / arc()	Draws the blue rectangle and red circle.
moveTo() / lineTo() / stroke()	Draws the green line.
fillText()	Displays the text "vijayasri" on the canvas.
footer	Creates the styled footer at the bottom of the page.

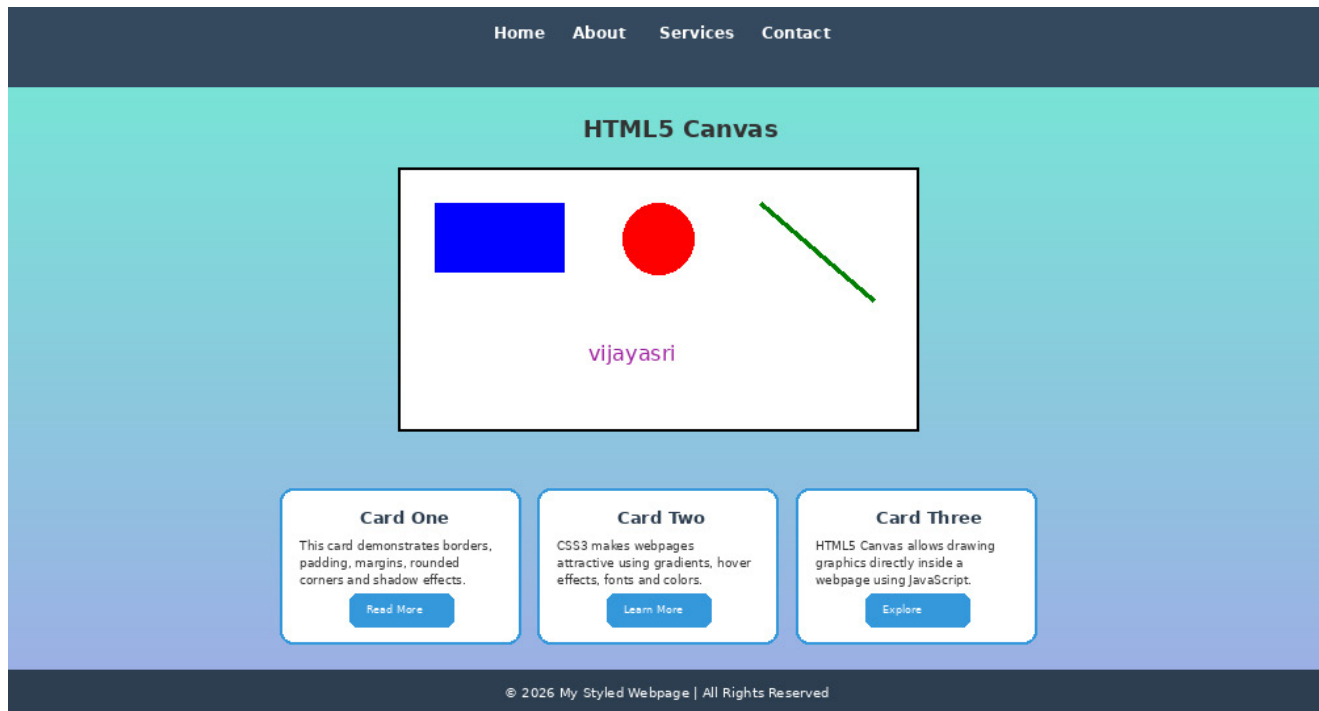
8. TEST CASES AND EXECUTION DATA

The following observations are based on the provided repository output screenshots. No unshown execution result has been invented.

Test Case	Input / Feature	Expected Result	Actual Result	Status
TC-01	Canvas drawing	Rectangle, circle, line and text should appear	All four canvas drawings are visible in the supplied output	Observed
TC-02	Three cards	Cards should be arranged and styled	Three bordered, rounded cards with shadows are visible	Observed
TC-03	Buttons	Each card should contain a styled button	Read More, Learn More and Explore buttons are visible	Observed
TC-04	Page styling	Gradient background, navigation and footer should render	The supplied output shows the gradient page, navigation and footer	Observed

9. OUTPUT

Output 1: HTML5 Canvas, styled cards and footer



10. RESULT

Thus, the styled webpage was developed using HTML5, CSS3 and JavaScript. Canvas graphics, CSS3 styling, three information cards, buttons, navigation and a footer were implemented and verified against the supplied repository output screenshot.
